

PERFORMANCE AUDIT REPORT

YELLOW TAXI - OCTOBER 2025

Subject Dataset: Taxi Fare Base Audit

CLEAN DATA & RETRAIN

DATA QUALITY SCORE: 97.34 (HIGH)

CLEAN DATA & RETRAIN

*** AUDIT RISK PROFILE ***

EXTREME SKEW: MODEL MAY OVER-PREDICT.
FAT-TAILS: OUTLIERS EXERT HIGH LEVERAGE ON PERFORMANCE STATS.
CONCENTRATED RISK: 72% OF TOP ERRORS SHARE PASSENGER_COUNT='1'.

Winning Model: Random Forest Regressor
Validation Score (r2): 0.7635
Test Score (r2): 0.6817
Throughput: 5.4K rows/sec
Integrity: 97.34/100 Score

Table of Contents

Executive Summary 1

Data Anomaly Log 3

Legend / Glossary 4

High-Level Competition Results 5

Generalization (CV vs Full) 6

Feature List 7

Random Forest Regressor [1] 8

Decision Tree Regressor [2] 9

Linear Regression [3] 10

Elastic Net [4] 11

Ridge Regression [5] 12

Lasso Regression [6] 13

Detailed Audit Stats 14

Recommendation 15

Data Anomaly Log: Random Forest Regressor

Top 15 Primary Contributors to a Dataset Kurtosis of 44.02 (out of 3.54M)

Actual	Predicted	Abs Error	Distance	Passengers	DO Hr	PU Hr	PU Day	DO Day
\$ 800.00	\$ 10.43	789.5699	0.02 mi	1	3	3	0	0
\$ 792.85	\$ 89.66	703.1880	0.00 mi	1	16	16	0	0
\$ -618.30	\$ -77.75	540.5527	96.30 mi	1	20	17	2	2
\$ -480.00	\$ 40.95	520.9451	0.00 mi	1	18	18	6	6
\$ -347.40	\$ 153.20	500.5980	53.75 mi	1	13	10	5	5
\$ -500.00	\$ -8.29	491.7112	0.00 mi	2	20	20	6	6
\$ 661.00	\$ 179.47	481.5286	28.45 mi	1	10	10	4	4
\$ 480.00	\$ -0.62	480.6218	0.00 mi	1	15	15	1	1
\$ -473.00	\$ 2.99	475.9921	0.00 mi	1	21	21	6	6
\$ 500.00	\$ 26.95	473.0486	0.00 mi	1	15	15	6	6
\$ 450.40	\$ -13.61	464.0088	0.00 mi	2	9	9	0	0
\$ -450.40	\$ -13.61	436.7912	0.00 mi	2	9	9	0	0
\$ 600.00	\$ 163.50	436.4967	0.88 mi	1	20	20	3	3
\$ -374.70	\$ 54.29	428.9911	63.19 mi	1	19	17	2	2
\$ 500.00	\$ 93.85	406.1523	0.41 mi	1	23	23	5	5

Model Audit Legend

--- Regression ---

-  DTR: Decision Tree Regressor
-  ENT: Elastic Net
-  LAS: Lasso Regression
-  LIN: Linear Regression
-  RFR: Random Forest Regressor
-  RGE: Ridge Regression

--- Classification ---

-  DTC: Decision Tree Classifier
-  LOG: Logistic Regression
-  RFC: Random Forest Classifier
- Unknown ---
-  UNK: Unknown

Note: Metric 'Score' refers to R^2 for Regression and Accuracy for Classification.

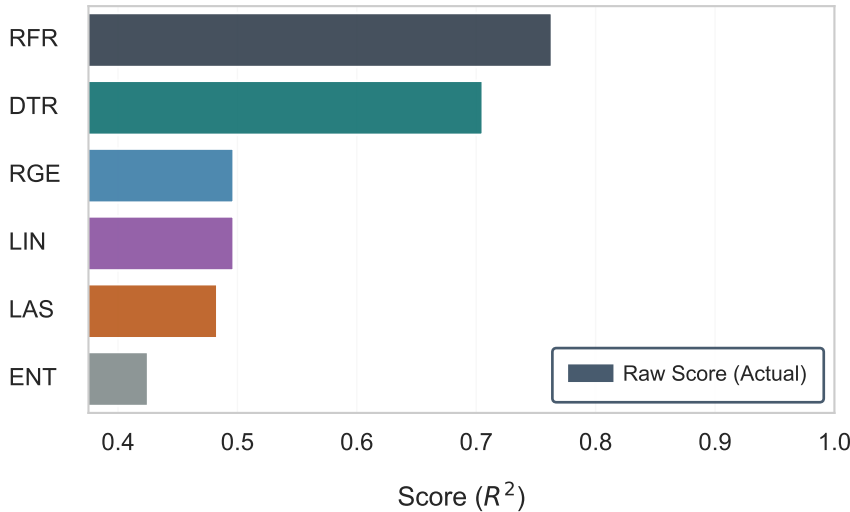
YELLOW TAXI - OCTOBER 2025

Dataset: Taxi Fare Base Audit

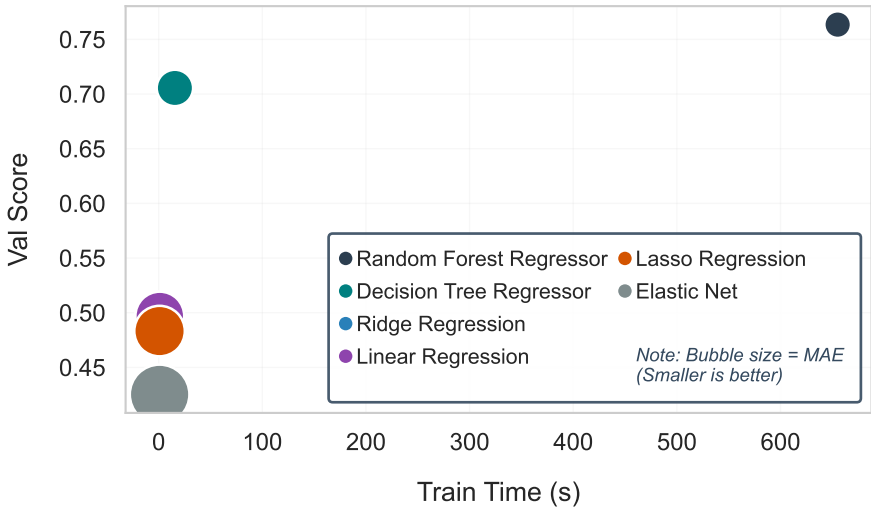
HIGH-LEVEL COMPETITION RESULTS

Page 5

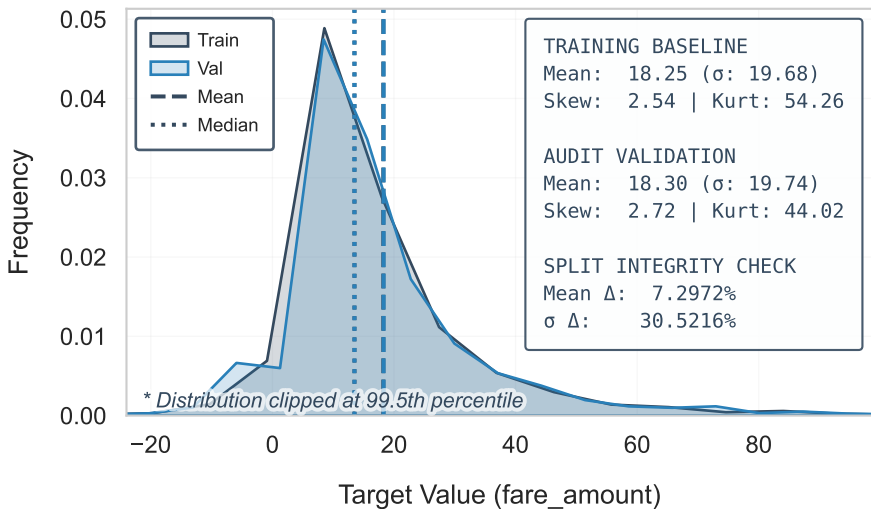
Predictive Accuracy (R^2) - Baseline Performance



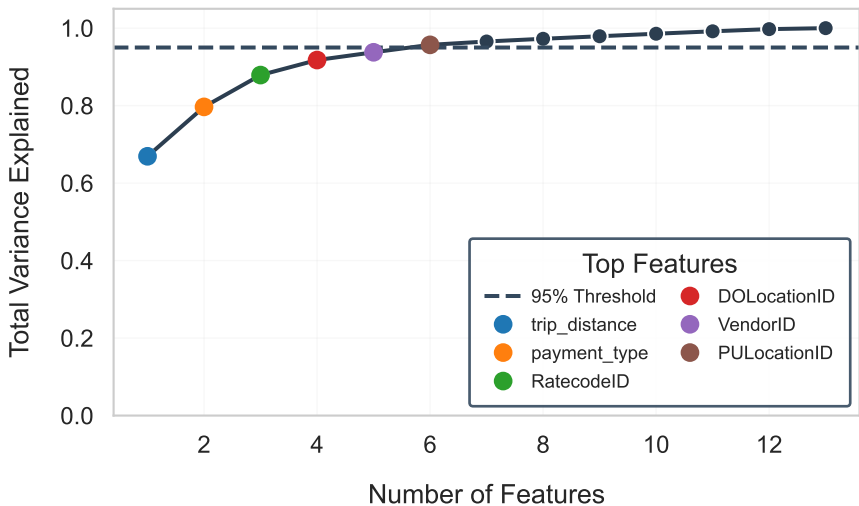
Efficiency: Accuracy vs. Time



Target Distribution & Statistical Shape



Cumulative Importance (Random Forest Regressor)



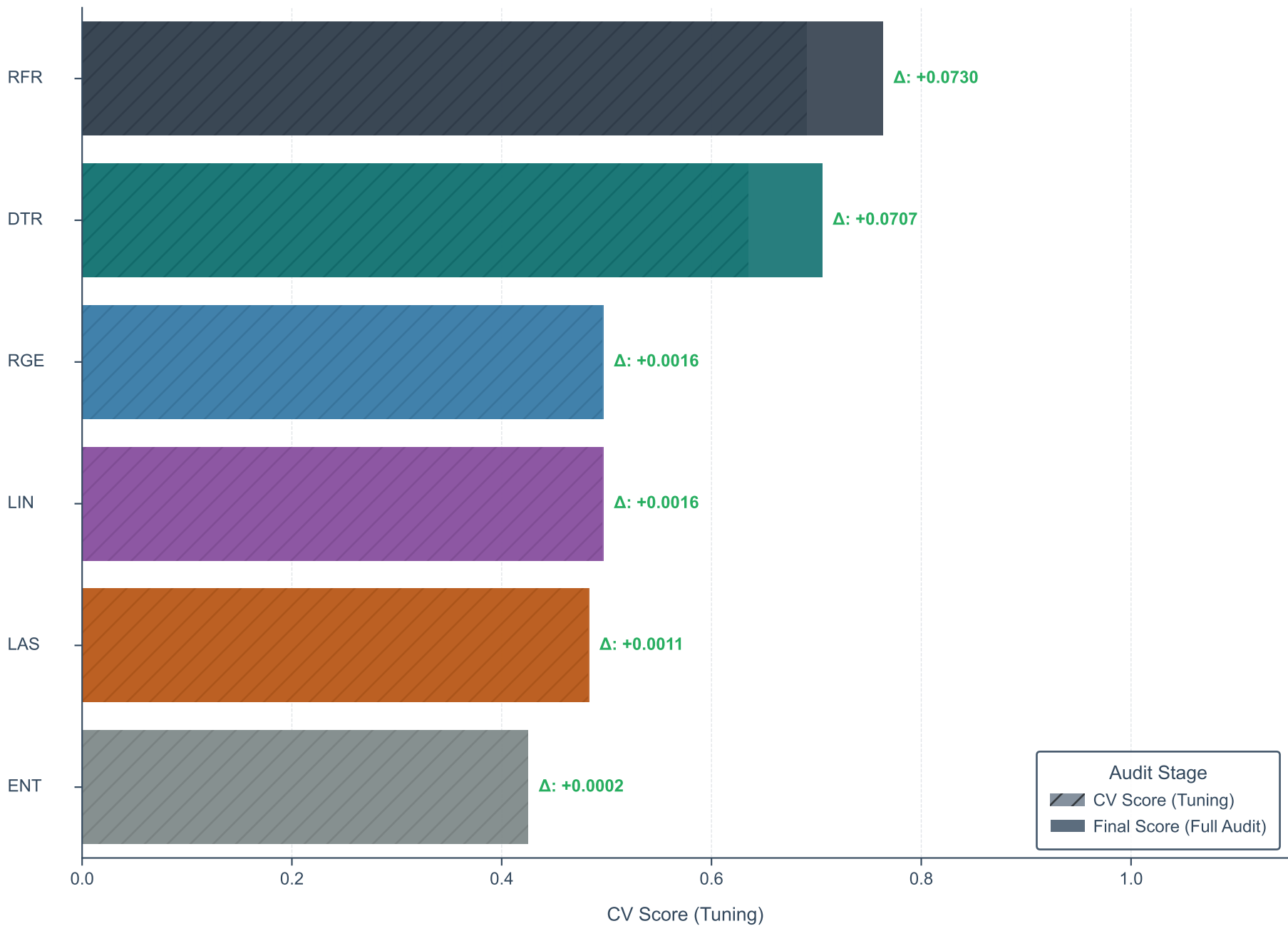
Data Integrity Check: SAFE (Drift: 0.275%)

YELLOW TAXI - OCTOBER 2025

Dataset: Taxi Fare Base Audit

GENERALIZATION (CV VS FULL)

Page 6



Features

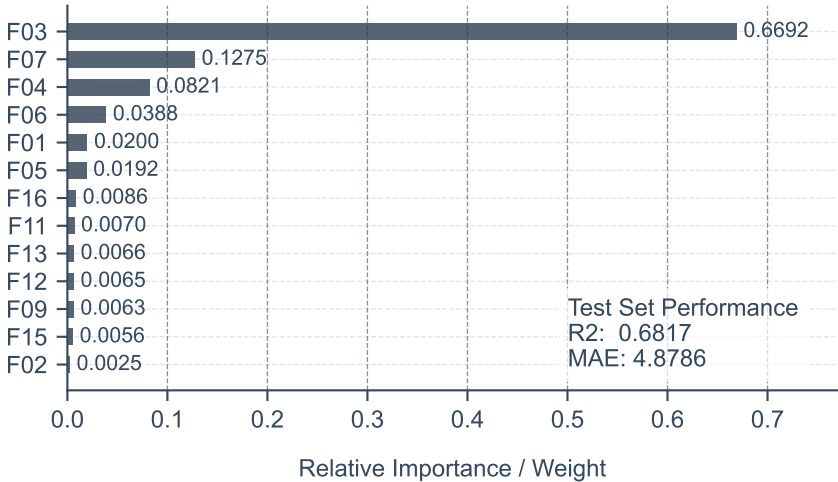
ID	Feature	Pos	Short Name	Parent Name	Used in Fit	Description
F01	VendorID	0	VendorID		Yes	
F02	passenger_count	1	Passengers		Yes	
F03	trip_distance	2	Distance		Yes	
F04	RatecodeID	3	Rate		Yes	
F05	PULocationID	4	PU Loc		Yes	
F06	DOLocationID	5	DO Loc		Yes	
F07	payment_type	6	Pmt Type		Yes	
F08	fare_amount	7	Fare		Yes	
F09	dropoff_dayofweek	8	DO Day		Yes	
F10	dropoff_hour	9	DO Hr		No	
F11	dropoff_hour_sin	10	DO Hr	dropoff_hour	Yes	
F12	dropoff_hour_cos	11	DO Hr	dropoff_hour	Yes	
F13	pickup_dayofweek	12	PU Day		Yes	
F14	pickup_hour	13	PU Hr		No	
F15	pickup_hour_sin	14	PU Hr	pickup_hour	Yes	
F16	pickup_hour_cos	15	PU Hr	pickup_hour	Yes	

Random Forest Regressor [1]

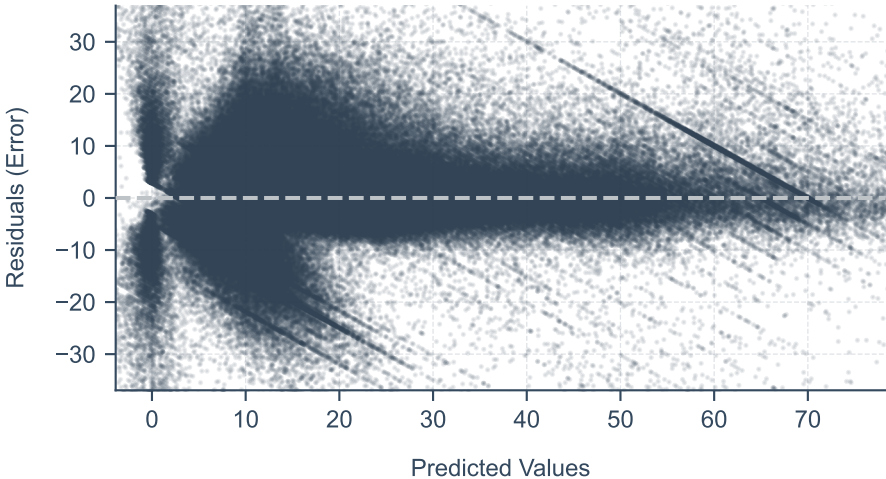
Configuration

HP	Value	HP	Value
Seed	75	Class Wts	None
Opt. Strategy	Random Search		
Criterion	squared_error		
Max Depth	20		
Trees	50		
Min Split	2		
Min Leaf	5		
Max Feat.	1.0		
Bootstrap	True		
Task	Regression		
Metric	r2		

Top 13 Predictors



Residual Analysis



Worst Regression Misses (Outliers)

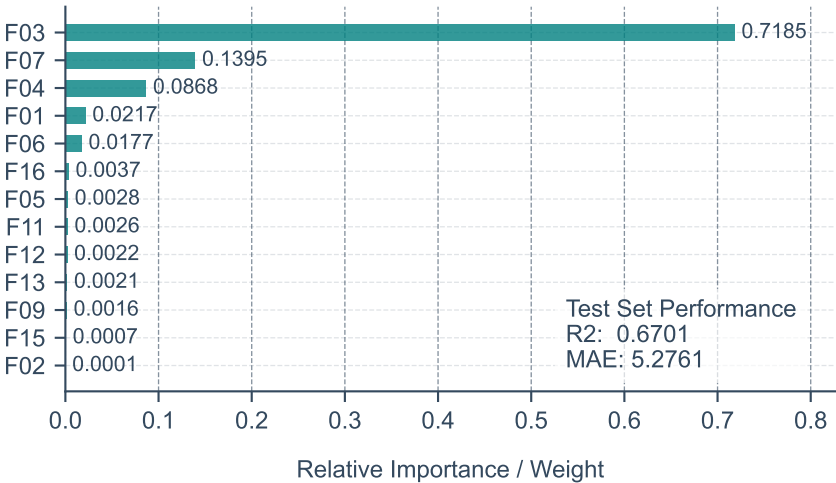
Actual	Predicted	Abs Error	Error Pct
\$ 800.00	\$ 10.43	789.5699	98.7%
\$ 792.85	\$ 89.66	703.1880	88.7%
\$ -618.30	\$ -77.75	540.5527	-87.4%
\$ -480.00	\$ 40.95	520.9451	-108.5%
\$ -347.40	\$ 153.20	500.5980	-144.1%

Decision Tree Regressor [2]

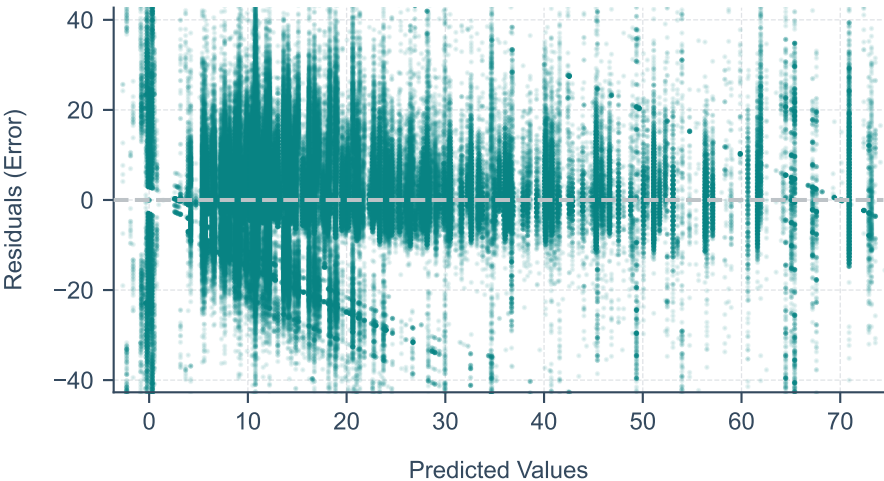
Configuration

HP	Value	HP	Value
Seed	75	Class Wts	None
Opt. Strategy	Random Search	Ccp Alpha	0.0
Criterion	squared_error	Task	Regression
Splitter	best	Metric	r2
Max Depth	10		
Min Split	2		
Min Leaf	1		
Min Wt Leaf	0.0		
Max Feat.	None		
Max Nodes	None		
Min Impurity	0.0		

Top 13 Predictors



Residual Analysis



Worst Regression Misses (Outliers)

Actual	Predicted	Abs Error	Error Pct
\$ 800.00	\$ -21.27	821.2697	102.7%
\$ -347.40	\$ 262.24	609.6422	-175.5%
\$ 661.00	\$ 95.37	565.6311	85.6%
\$ 600.00	\$ 34.69	565.3095	94.2%
\$ 500.00	\$ -0.34	500.3390	100.1%

YELLOW TAXI - OCTOBER 2025

Dataset: Taxi Fare Base Audit

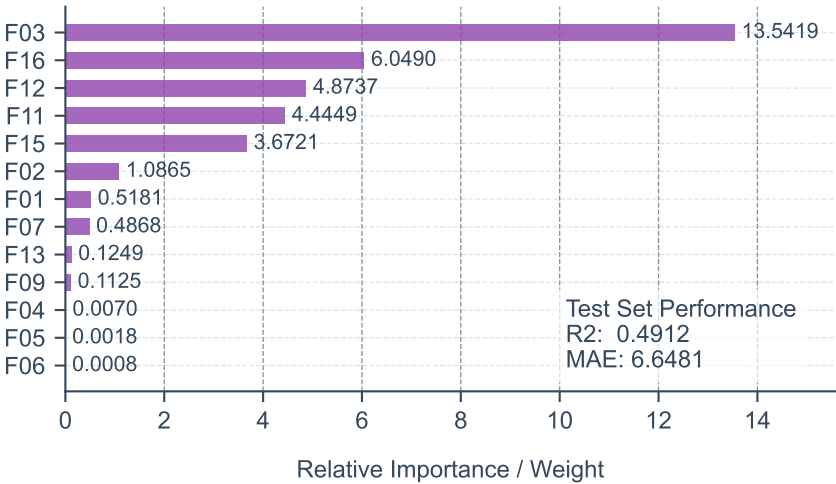
Page 10

Linear Regression [3]

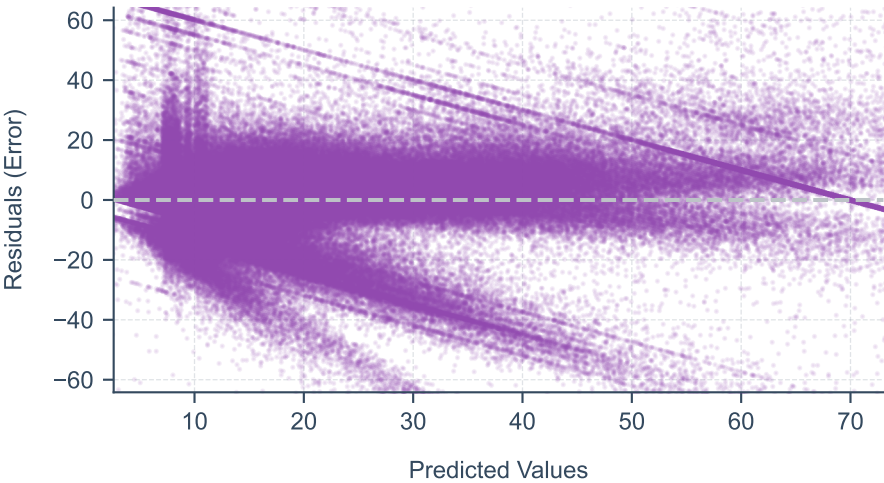
Configuration

HP	Value
Seed	1
Opt. Strategy	Manual
Intercept	True
Copy X	True
N Jobs	None
Positive	False
Task	Regression
Metric	r2

Top 13 Predictors



Residual Analysis



Worst Regression Misses (Outliers)

Actual	Predicted	Abs Error	Error Pct
\$ -618.30	\$ 308.20	926.4993	-149.8%
\$ 800.00	\$ 3.19	796.8140	99.6%
\$ 792.85	\$ 7.04	785.8130	99.1%
\$ 600.00	\$ 8.59	591.4058	98.6%
\$ -374.70	\$ 205.52	580.2234	-154.9%

YELLOW TAXI - OCTOBER 2025

Dataset: Taxi Fare Base Audit

Page 11

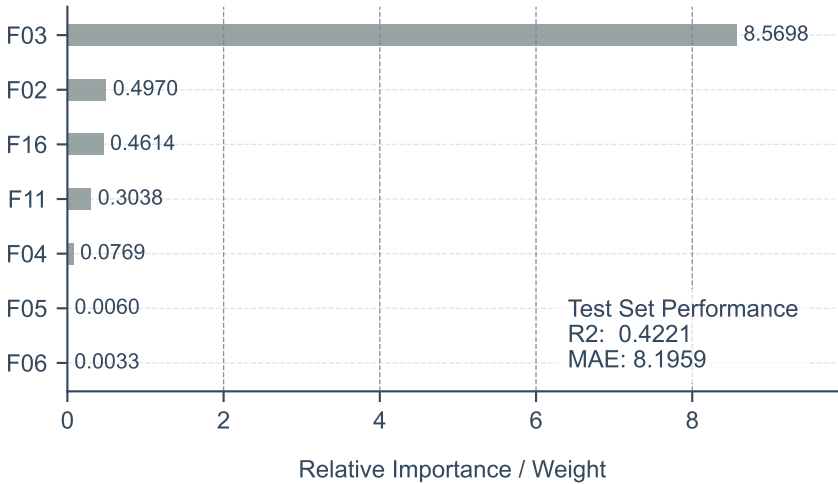
Elastic Net [4]

Configuration

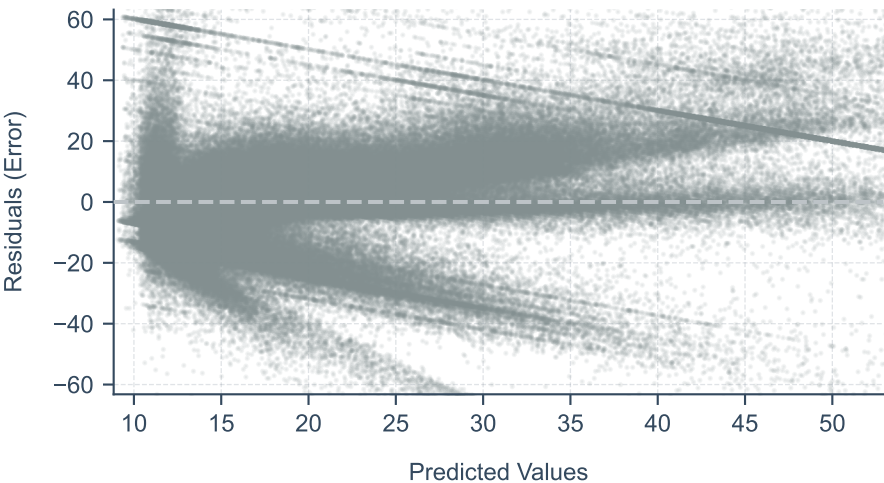
HP	Value
Seed	1
Opt. Strategy	Manual
Alpha	1.0
L1 Ratio	0.5
Intercept	True
Copy X	True
Max Iter	1000
Tol	0.0001
Warm Start	False
Positive	False
Selection	cyclic

HP	Value
Task	Regression
Metric	r2

Top 7 Predictors



Residual Analysis



Worst Regression Misses (Outliers)

Actual	Predicted	Abs Error	Error Pct
\$ -618.30	\$ 199.56	817.8618	-132.3%
\$ 800.00	\$ 9.95	790.0538	98.8%
\$ 792.85	\$ 12.52	780.3253	98.4%
\$ 661.00	\$ 66.46	594.5416	89.9%
\$ 600.00	\$ 12.94	587.0625	97.8%

YELLOW TAXI - OCTOBER 2025

Dataset: Taxi Fare Base Audit

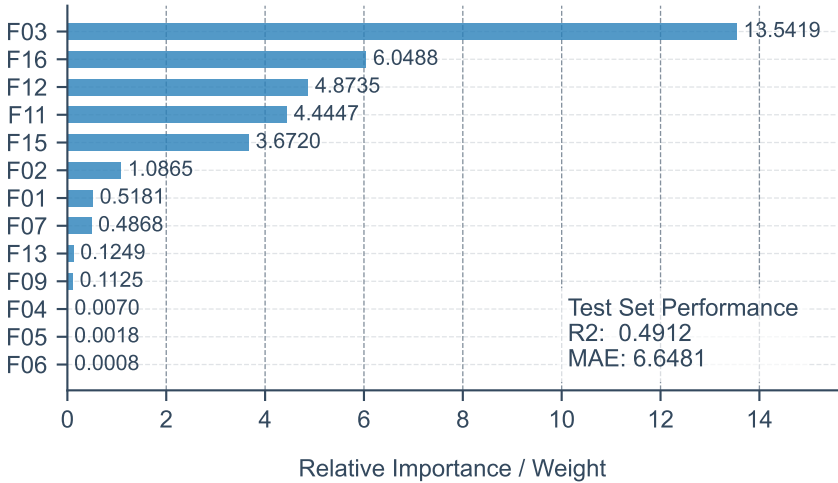
Page 12

Ridge Regression [5]

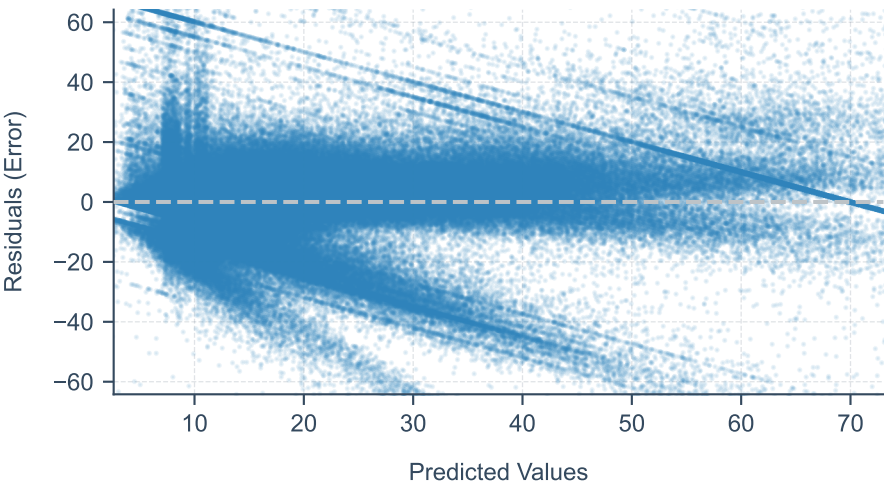
Configuration

HP	Value
Seed	1
Opt. Strategy	Manual
Alpha	1.0
Intercept	True
Copy X	True
Max Iter	None
Tol	0.0001
Solver	auto
Task	Regression
Metric	r2

Top 13 Predictors



Residual Analysis



Worst Regression Misses (Outliers)

Actual	Predicted	Abs Error	Error Pct
\$ -618.30	\$ 308.20	926.4992	-149.8%
\$ 800.00	\$ 3.19	796.8140	99.6%
\$ 792.85	\$ 7.04	785.8129	99.1%
\$ 600.00	\$ 8.59	591.4058	98.6%
\$ -374.70	\$ 205.52	580.2233	-154.9%

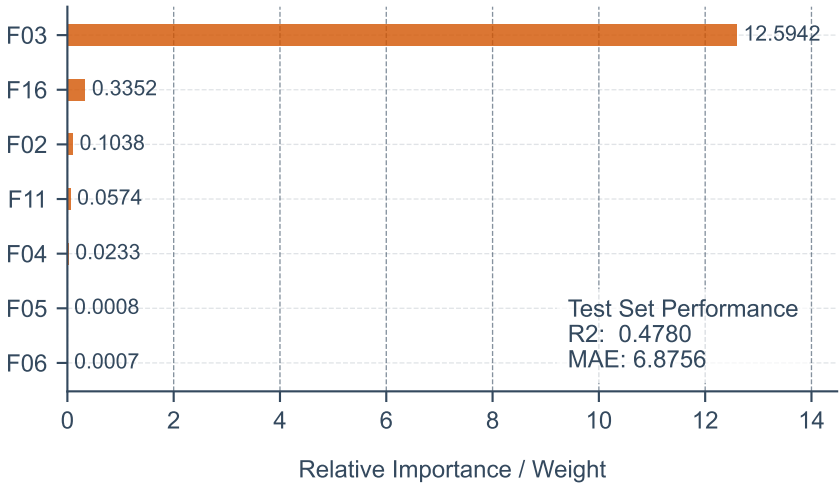
Lasso Regression [6]

Configuration

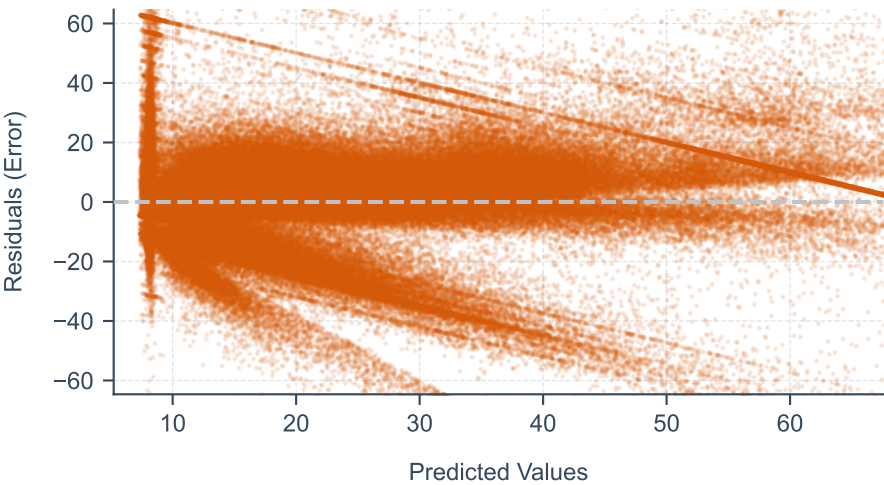
HP	Value
Seed	1
Opt. Strategy	Manual
Alpha	1.0
Intercept	True
Copy X	True
Precompute	False
Max Iter	1000
Tol	0.0001
Warm Start	False
Positive	False
Selection	cyclic

HP	Value
Task	Regression
Metric	r2

Top 7 Predictors



Residual Analysis



Worst Regression Misses (Outliers)

Actual	Predicted	Abs Error	Error Pct
\$ -618.30	\$ 284.74	903.0358	-146.1%
\$ 800.00	\$ 7.86	792.1427	99.0%
\$ 792.85	\$ 8.35	784.5045	98.9%
\$ 600.00	\$ 10.51	589.4885	98.2%
\$ 661.00	\$ 90.12	570.8830	86.4%

Detailed Audit Stats - 20260208_1600

Model	Abbr	CV Score (Tuning)	Val Score	Test Score	Train Time (s)	Actual Peak RAM
Random Forest Regressor	RFR	0.6905	0.7635	0.6817	655.14	2.54GB
Decision Tree Regressor	DTR	0.6348	0.7056	0.6701	15.67	3.69GB
Ridge Regression	RGE	0.4954	0.4969	0.4912	0.83	4.13GB
Linear Regression	LIN	0.4954	0.4969	0.4912	1.12	4.12GB
Lasso Regression	LAS	0.4821	0.4832	0.4780	0.86	4.14GB
Elastic Net	ENT	0.4251	0.4253	0.4221	0.92	4.13GB

STRATEGIC RECOMMENDATION

Winning Model: **Random Forest Regressor**
Validation Score (r2): **0.7635**
Test Score (r2): **0.6817**
Throughput: **5.4K rows/sec**
Integrity: **97.34/100 Score**
Audit Scale: **3.54M processed**
Audit Duration: **11m 14s**
Resources: **10 Cores | Peak Mem: 4.14GB**
Methodology: **14.1% used for training**

NOTES

- Significant score improvement on full data vs sample.
- This model required aggressive sampling to stay within hardware memory limits during the tuning phase.

This recommendation is based on a balance of validation accuracy, memory efficiency, and training time performance.